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Roll No

AU/ME-703-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Mechanical Vibration and Noise Engineering

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is the vibration and explain about it's main causes?
Define the phenomena of resonance.
b) An unknown mass m is attached to one end of a spring of stiffness k having natural frequency of 6Hz. When 1 kg mass is attached with m the natural frequency of the system is lowered by 20%. Determine the value of known mass m and stiffness k .
2. What is the difference between deterministic and random vibration? Give two practical examples of each. What are the common types of damping?
3. Explain in brief, following terms with respect to noise:
i) Loudness and equal loudness contours
ii) Auditory effect of noise
4. Derive an expression for frequency of damped vibrations for a spring mass system with the mass capable of sliding on a dry surface.

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5. a) Explain the following terms:
 - i) Sound spectra
 - ii) Octave band analysis
- b) Explain about the major sources of noise on road and in industries.
6. Derive the equation for natural frequency of vibration of compound pendulum.
7. Write the equations of motion of a multi-degree-of-freedom system in matrix form using:
 - i) The flexibility matrix
 - ii) The stiffness matrix
8. Write short notes on any two:
 - a) The compound pendulum
 - b) Vibration isolators
 - c) Industrial noise control
